MQGT-SCF Phase II:

Minimal Scalar-Singlet EFT, Local GKSL Measurement Dynamics, and an H2 Interferometric Exclusion Observable

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Abstract

We present a compact Phase II formulation of the MQGT-SCF program as a minimal scalar-singlet effective field theory extending General Relativity and the Standard Model by two real singlets $\Phi_c(x)$ and $E(x)$. The technical claim is deliberately limited to *specification-level closure*: the framework fixes the action, field equations, decoupling limit, boundedness conditions, a mediator-derived nonlocal-kernel route, and a local covariant Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) measurement container compatible with completely positive trace-preserving evolution. Any teleology-like effect is retained only as an optional measure-level tilt rather than as a bulk time-odd source. We state a finite-dimensional operational no-signalling result for the safe local class and identify the main empirical target as excess interferometric dephasing in the H2 channel, summarized by a visibility-suppression law and an instrument-bounded exclusion floor. A controlled low-energy naturalness module is added by adopting an EFT cutoff benchmark $\Lambda_\star = 1\,\mathrm{TeV}$ with stress band $\Lambda \in [1, 10]\,\mathrm{TeV}$, which separates an ultralight mainline sector from a heavier collider-safe companion benchmark. The result is a smaller and more auditable object than an all-at-once unification claim: a framework that can be constrained or falsified in clear ways.

1 Introduction

Ambitious unification programs often fail rhetorically before they fail physically. When field content, measurement deformation, teleology, phenomenology, agent architecture, and implementation claims are presented as coequal from the outset, the mathematically explicit core is easily obscured. The present paper adopts a stricter posture.

The goal here is not to claim final empirical confirmation, nor to restate the full philosophical scope of the wider MQGT-SCF corpus. The goal is to state the smallest mathematically coherent and empirically auditable core currently available. We use the phrase *specification-level closure* to mean that the mathematical objects are explicit, the limiting cases are explicit, and the parameter inventory is finite and operationally linked to data.

The framework is organized around five coupled elements: (i) a minimal scalar-singlet EFT extending General Relativity and the Standard Model by two real singlets $\Phi_c(x)$ and $E(x)$; (ii) the corresponding classical field equations together with a mediator-derived route to an effective nonlocal kernel; (iii) a local covariant GKSL measurement container; (iv) an optional measure tilt retained as a companion statistical layer rather than a bulk time-odd teleology source; and (v) a flagship interferometric exclusion observable for the H2 channel.

A further ingredient added here is a controlled naturalness analysis for the scalar-singlet sector. Rather than claiming a high-scale ultraviolet completion, we adopt a deliberately low-energy EFT stance and require technical control only within a finite cutoff window. This sharpens the viable parameter region and makes the connection to experiment more disciplined.

2 Minimal scalar-singlet EFT core

We work on a four-dimensional Lorentzian spacetime $(\mathcal{M}, g_{\mu\nu})$ with natural units $\hbar = c = 1$. The minimal bulk is General Relativity plus the Standard Model plus two real gauge-singlet scalar sectors, denoted $\Phi_c(x)$ and $E(x)$. The anchor action is

$$S_{\text{core}} = \int_{\mathcal{M}} d^4x \sqrt{-g}\,(\mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SM}} + \mathcal{L}_{\Phi} + \mathcal{L}_{E} + \mathcal{L}_{\text{int}}). \tag{1}$$

The gravitational and singlet sectors are

$$\mathcal{L}_{\text{GR}} = \frac{M_{\text{Pl}}^2}{2} R - \Lambda, \tag{2}$$

$$\mathcal{L}_{\Phi} = -\frac{1}{2}(\nabla\Phi_c)^2 - V_{\Phi}(\Phi_c), \tag{3}$$

$$\mathcal{L}_{E} = -\frac{1}{2}(\nabla E)^2 - V_{E}(E), \tag{4}$$

with renormalizable potentials

$$V_{\Phi}(\Phi_c) = \frac{1}{2} m_{\Phi}^2 \Phi_c^2 + \frac{\lambda_{\Phi}}{4}\Phi_c^4, \tag{5}$$

$$V_{E}(E) = \frac{1}{2} m_{E}^2 E^2 + \frac{\lambda_{E}}{4} E^4. \tag{6}$$

A minimal interaction sector is

$$\mathcal{L}_{\text{int}} = -\frac{\kappa}{2}\Phi_c^2 E^2 - \gamma \Phi_c E - \frac{\lambda_{h\Phi}}{2}\Phi_c^2 H^\dagger H - \frac{\lambda_{hE}}{2} E^2 H^\dagger H + J_{\Phi}\Phi_c + J_E E, \tag{7}$$

where H is the Standard Model Higgs doublet and $J_{\Phi}(x), J_E(x)$ denote external or effective source functionals. The interpretive labels "consciousness field" and "ethical field" are retained only for continuity; mathematically the construction is a two-singlet EFT coupled to gravity and the Standard Model through renormalizable portals and effective sources.

Proposition 1 (Bounded quartic sector)**.** *The quartic potential*

$$V_4(\Phi_c, E) = \frac{\lambda_{\Phi}}{4}\Phi_c^4 + \frac{\lambda_E}{4} E^4 + \frac{\kappa}{2}\Phi_c^2 E^2 \tag{8}$$

is bounded from below if

$$\lambda_\Phi > 0, \qquad \lambda_E > 0, \qquad \kappa > -\sqrt{\lambda_\Phi \lambda_E}. \tag{9}$$

Variation of Eq. (1) gives the Einstein equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{2}{M_{\rm Pl}^2} T_{\mu\nu}^{\rm total}, \tag{10}$$

and the singlet field equations

$$\Box\Phi_c + m_\Phi^2 \Phi_c + \lambda_\Phi \Phi_c^3 + \kappa \Phi_c E^2 + \gamma E + \lambda_{h\Phi} \Phi_c H^\dagger H = J_\Phi, \tag{11}$$

$$\Box E + m_E^2 E + \lambda_E E^3 + \kappa E \Phi_c^2 + \gamma \Phi_c + \lambda_{hE} E H^\dagger H = J_E. \tag{12}$$

A useful closure route for any effective nonlocal kernel is to derive it by integrating out a UV-local mediator. Introduce an auxiliary real scalar $M(x)$ with action

$$S_M = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu M \partial_\nu M - \frac{1}{2} m_M^2 M^2 - g_\Phi M \Phi_c - g_E M E \right]. \tag{13}$$

Integrating out M yields the effective nonlocal interaction

$$S_{\rm nonloc}[\Phi_c, E] = \frac{1}{2} \int d^4x \sqrt{-g} \int d^4x' \sqrt{-g'}\, J(x) G(x,x') J(x'), \tag{14}$$

with $J(x) = g_\Phi \Phi_c(x) + g_E E(x)$ and kernel

$$K(x,x') = g_\Phi g_E\, G(x,x'). \tag{15}$$

In the quasi-static Minkowski limit, $K(r) \propto e^{-m_M r}/(4\pi r)$, while for $m_M \to \infty$ one recovers a local derivative expansion.

Remark 1 (Decoupling)**.** *If the novel couplings are taken to zero and the singlet sectors are absent or dynamically frozen,*

$$\kappa, \gamma, \lambda_{h\Phi}, \lambda_{hE}, J_\Phi, J_E \to 0, \tag{16}$$

then the framework reduces to GR+SM at low energy. This is the essential consistency requirement for the anchor core.

3 Local GKSL measurement container and optional measure tilt

The measurement layer is formulated as a local covariant open-system channel written in GKSL form. In a Tomonaga-Schwinger hypersurface description, the density operator evolves according to

$$\frac{\delta\rho[\Sigma]}{\delta\Sigma(x)} = -i[H(x), \rho[\Sigma]] + \sum_a \left(L_a(x) \rho[\Sigma] L_a^\dagger(x) - \frac{1}{2} \{ L_a^\dagger(x) L_a(x), \rho[\Sigma] \} \right). \tag{17}$$

A minimal local modulation is

$$L_a(x) = \sqrt{\lambda_a(E(x), \Phi_c(x))}\, \widehat{M}_a(x), \qquad \lambda_a(E, \Phi_c) = \lambda_{a,0} e^{\eta_a E}, \qquad |\eta_a E| \ll 1, \tag{18}$$

with nonnegative baseline rates $\lambda_{a,0} \geq 0$ and local measurement operators $\widehat{M}_a(x)$. As long as the rates remain nonnegative, the evolution is completely positive and trace preserving.

At the effective outcome level, a corresponding normalized exponential tilt is

$$P_\eta(i) = \frac{|c_i|^2 e^{\eta \Delta E_i}}{\sum_j |c_j|^2 e^{\eta \Delta E_j}}, \qquad |\eta| \ll 1, \tag{19}$$

which linearizes to

$$P_\eta(i) = |c_i|^2 \Big[1 + \eta(\Delta E_i - \langle \Delta E \rangle)\Big] + \mathcal{O}(\eta^2). \tag{20}$$

The crucial operational point is that any measurement deformation is confined to a local, linear, CPTP container. This is the smallest disciplined setting in which the no-signalling question can be posed cleanly.

4 Controlled EFT regime and benchmark parameter card

A central consistency requirement is that the scalar-singlet sector remain technically controlled in the presence of Standard Model radiative corrections. The present formulation therefore adopts a low-energy EFT posture rather than a high-scale UV-completion claim.

We define a central cutoff benchmark

$$\Lambda_\star = 1\,\text{TeV}, \tag{21}$$

with a stress-test band

$$\Lambda \in [1, 10]\ \text{TeV}. \tag{22}$$

Naturalness is enforced only within this EFT validity window.

The dominant one-loop radiative corrections to the singlet masses arise from the Higgs-portal terms:

$$\delta m_\Phi^2 \sim \frac{\lambda_{h\Phi}}{16\pi^2}\Lambda^2, \qquad \delta m_E^2 \sim \frac{\lambda_{hE}}{16\pi^2}\Lambda^2. \tag{23}$$

Imposing the technical-naturalness conditions

$$|\delta m_i^2| \lesssim \mathcal{O}(m_i^2), \qquad i \in \{\Phi, E\}, \tag{24}$$

implies the portal ceilings

$$\lambda_{hs} \lesssim 16\pi^2 \frac{m_s^2}{\Lambda^2}, \qquad s \in \{\Phi, E\}. \tag{25}$$

This immediately separates the viable region into two operational sectors. In an ultralight mainline sector with $m_s \sim 10^{-4}$–10^{-3} eV, relevant to coherence-facing and sub-millimeter force searches, the portal ceilings at $\Lambda_\star = 1\,\text{TeV}$ are $\lambda_{hs} \lesssim 1.6 \times 10^{-30}$–$1.6 \times 10^{-28}$. In a heavier companion sector with $m_s \sim 1$–$10\,\text{GeV}$, the corresponding ceiling is $\lambda_{hs} \lesssim 1.6 \times 10^{-4}$–$1.6 \times 10^{-2}$.

The local GKSL container also matters structurally: it narrows the allowed deformation class and excludes an entire family of uncontrolled nonlocal amplification channels. In this way, naturalness is achieved through layered suppression: small Higgs-portal couplings, an explicit low EFT cutoff, confinement of measurement dynamics to a local CPTP container, and retention of teleology only

as an optional measure tilt rather than a bulk time-odd source.

The benchmark parameter card used in the present paper is given in Table 1.

Parameter	Meaning	Central benchmark	Allowed / target regime	Status
Λ	EFT cutoff	1 TeV	1–10 TeV	fixed benchmark
m_Φ	Φ_c physical mass	10^{-4}–10^{-3} eV	ultralight mainline	fit target / benchmark
m_E	E physical mass	10^{-4}–10^{-3} eV	ultralight mainline	fit target / benchmark
$\lambda_{h\Phi}$	Higgs–Φ_c portal	$\lesssim 1.6 \times 10^{-30}$ to 1.6×10^{-28}	$\ll 1$	naturalness bound
λ_{hE}	Higgs–E portal	$\lesssim 1.6 \times 10^{-30}$ to 1.6×10^{-28}	$\ll 1$	naturalness bound
$\sin^2\theta_{h\Phi}$	Higgs mixing for Φ_c	effectively negligible	$\ll 1$	derived
$\sin^2\theta_{hE}$	Higgs mixing for E	effectively negligible	$\ll 1$	derived
ξ	optional measure-level teleological tilt	≈ 0 operationally	$\lvert\xi\rvert \ll 1$	companion-only / tiny
η	measure-tilt parameter	≈ 0 operationally	$\lvert\eta\rvert \ll 1$	companion-only / tiny
m_s^{comp}	companion singlet mass	1–10 GeV	heavier singlet sector	optional benchmark
$\lambda_{hs}^{\mathrm{comp}}$	portal at 1–10 GeV	$\lesssim 1.6 \times 10^{-4}$ to 1.6×10^{-2}	weak but nonzero	companion benchmark

Table 1: Benchmark card for the MQGT-SCF low-energy EFT used in the present Phase II formulation.

5 H2 interferometric exclusion observable

The flagship physical falsifier for the anchor program is interferometric visibility suppression in the H2 channel. The preregistered signal model is

$$\frac{V}{V_0} = \exp(-\Gamma T\,\Delta x^2), \tag{26}$$

where V is the measured interferometric visibility, V_0 is the baseline visibility predicted by the nuisance-only instrument model, T is the effective superposition time, Δx is the effective path separation, and Γ is an excess dephasing coefficient. Defining the sensitivity coordinate

$$Q := T\,\Delta x^2, \tag{27}$$

accepted sweeps are analyzed through the preregistered regression model

$$-\ln V_{jr} = a_{s(r)} + \Gamma Q_j + X_{jr}^\top \beta + \varepsilon_{jr}, \tag{28}$$

with session-specific intercepts and nuisance regressors.

The core deliverable of the Phase-0 pilot is not discovery language but the achieved exclusion floor

$$\Gamma_{\text{floor}} = \frac{|\ln(1-\delta_{\text{tot}})|}{T\,\Delta x^2}, \tag{29}$$

where δ_{tot} is the preregistered total nuisance budget. Using the reference profile

$$T_{\text{ref}} = 10^{-6}\,\text{s}, \qquad \Delta x_{\text{ref}} = 10^{-3}\,\text{m}, \qquad \delta_{\text{tot,ref}} = 1.15\times 10^{-3}, \tag{30}$$

one obtains

$$\Gamma_{\text{floor,ref}} \approx 1.15\times 10^{9}\,\text{s}^{-1}\text{m}^{-2}. \tag{31}$$

This is a planning benchmark, not a hardware discovery claim.

The key interpretive asymmetry of H2 is worth preserving. A stable null is a scientific success because it produces a real, instrument-bounded exclusion statement. A stable positive excess is not a discovery; it is only an escalation trigger for a stronger follow-up platform. That asymmetry is what makes the H2 channel compatible with the deliberately modest posture adopted here.

6 Conclusion

The MQGT-SCF program is strongest when presented not as a triumphant final answer, but as a closed research architecture with a mathematically explicit core. In the formulation given here, that core consists of a minimal scalar-singlet EFT action, explicit classical field equations, a mediator-derived kernel route, a local covariant GKSL measurement container, an optional measure-level tilt retained as a companion statistical layer, a finite operational no-signalling statement for the safe measurement class, a controlled low-energy naturalness module, and a direct bridge from parameters to a preregistered interferometric exclusion observable.

This is enough to justify a precise but limited claim: the program is closed at specification level. It is not enough to justify decisive empirical confirmation, a high-scale ultraviolet completion, or a uniqueness theorem for a final Theory of Everything. Those remain open. But because the symbols are fixed, the limiting cases are clear, and the experimental bridge is explicit, the remaining gaps are no longer hidden behind rhetoric. They have become ordinary scientific work.

A Compact benchmark logic

For convenience, the benchmark logic may be summarized as

$$\Lambda_\star = 1\,\text{TeV}, \qquad \Lambda \in [1,10]\ \text{TeV}, \tag{32}$$

$$\lambda_{hs}^{\max} \sim 16\pi^2 \frac{m_s^2}{\Lambda^2}, \tag{33}$$

$$m_s^{\text{main}} \sim 10^{-4}\text{–}10^{-3}\,\text{eV}, \tag{34}$$

$$m_s^{\text{comp}} \sim 1\text{–}10\,\text{GeV}. \tag{35}$$

The ultralight sector is the mainline anchor-paper regime. The heavier benchmark is included only to show that the broader corpus admits a collider-safe singlet realization without changing the

anchor argument.

B Operational no-signalling statement

Theorem 1 (Operational no-signalling under local factorization)**.** *Let $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ and suppose the joint state evolves according to*

$$\dot{\rho}_{AB}(t) = (\mathcal{L}_A^{u_A} \otimes \mathrm{id}_B + \mathrm{id}_A \otimes \mathcal{L}_B^{u_B})\rho_{AB}(t), \tag{36}$$

with each local generator of GKSL type and with explicitly local dependence on settings and source data. Then the reduced state on A satisfies

$$\dot{\rho}_A(t) = \mathcal{L}_A^{u_A}(\rho_A(t)), \qquad \rho_A(t) = \mathrm{Tr}_B \rho_{AB}(t), \tag{37}$$

so local outcome statistics in region A do not depend on free protocol choices made in spacelike-separated region B.

Proof sketch. Taking the partial trace over B gives

$$\dot{\rho}_A(t) = \mathrm{Tr}_B\big[(\mathcal{L}_A^{u_A} \otimes \mathrm{id}_B)\rho_{AB}(t)\big] + \mathrm{Tr}_B\big[(\mathrm{id}_A \otimes \mathcal{L}_B^{u_B})\rho_{AB}(t)\big]. \tag{38}$$

Because $\mathcal{L}_A^{u_A}$ acts only on A, the first term equals $\mathcal{L}_A^{u_A}(\rho_A)$. The second term vanishes because a GKSL generator is trace preserving on B. Hence $\dot{\rho}_A(t) = \mathcal{L}_A^{u_A}(\rho_A(t))$, independent of u_B. The statement does not cover nonlocal generators, nonlinear rules outside the GKSL container, or non-factorizing companion tilts. □

References

[1] G. Lindblad, "On the generators of quantum dynamical semigroups," *Communications in Mathematical Physics* **48** (1976), 119–130.

[2] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, "Completely positive dynamical semigroups of N-level systems," *Journal of Mathematical Physics* **17** (1976), 821–825.

[3] S. Tomonaga, "On a relativistically invariant formulation of the quantum theory of wave fields," *Progress of Theoretical Physics* **1** (1946), 27–42.

[4] J. Schwinger, "Quantum electrodynamics. I. A covariant formulation," *Physical Review* **74** (1948), 1439–1461.

[5] H. Georgi, "Effective field theory," *Annual Review of Nuclear and Particle Science* **43** (1993), 209–252.

[6] A. Bassi, K. Lochan, S. Satin, T. P. Singh, and H. Ulbricht, "Models of wave-function collapse, underlying theories, and experimental tests," *Reviews of Modern Physics* **85** (2013), 471–527.

[7] S. Nimmrichter, K. Hornberger, P. Haslinger, and M. Arndt, "Macroscopic matter-wave interferometry probes collapse-model parameters," *Physical Review Letters* **113** (2014), 020405.